TRANSIENT STABILITY OF A MULTI-MACHINE POWER SYSTEM
PART I : INVESTIGATION OF SYSTEM TRAJECTORIES

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Abstract — Direct methods of transient stability analysis of a multi-machine power system, using a function describing the system's transient energy, are discussed. By examining the trajectory of the disturbed system, the following fundamental questions are dealt with: the concept of a controlling unstable equilibrium point (u.e.p.), the manner in which some generators tend to lose synchronism, and identifying the energy directly responsible for system separation. Resolving these issues will substantially improve transient stability analysis by a direct method.

INTRODUCTION

Present day transient stability analysis of a multi-machine power system is mainly performed by simulations. For a given initial operating condition and a specified large disturbance or sequence of disturbances, a time solution is obtained for the generators' rotor angles, speeds, powers, terminal voltages, etc. By examining the generators' rotor angles at various instants, separation of one or more generators from the rest of the system, indicating loss of synchronism, is detected. While the magnitude of the computational effort involved depends on the complexity of the mathematical model used, only numerical methods can be used to obtain the time solution. Even for a small power network, and with the simplest mathematical model possible, this method is slow and cumbersome.

For many years there has been a great deal of interest in using direct methods for transient stability analysis (see [1] for a comprehensive review). These methods have in common the following general approach:

- Development of a special function by which the stability characteristics of the system's post disturbance equilibrium point is examined.
- Determination of the region of stability. This has the following practical significance: if at the beginning of the last transient (e.g., at the instant of fault clearing) the system trajectory is located inside that region, the system will be stable.

Both of the above areas have received a great deal of attention by investigators. For the first area, early investigators used functions that described the system energy [1,2,3]. Later, Lyapunov-type functions were suggested, and more recently, energy-type functions have been used [4,5]. The second area has been recognized as the major cause for the conservative results obtained with direct methods. In early work, the region of stability was determined by considering the unstable equilibrium point (u.e.p.) nearest to the stable equilibrium point. Later efforts have determined the appropriate u.e.p. for the particular disturbance under consideration [6,7,4].

There have been significant improvements in both areas discussed above. The state of the art is progressing toward achieving fairly accurate and reliable results, at least with simple power system models. When this is accomplished, the stage would be set for the use of direct methods of stability analysis in such applications as preliminary planning studies, and in some on-line applications, such as dynamic security assessment. For this to take place, however, some fundamental questions must be resolved. These questions pertain to the controlling u.e.p. for a particular system transient, and its relation to the manner in which some generators tend to lose synchronism.

To resolve these fundamental questions, we must understand the physical behavior of synchronous machines when subjected to large disturbances, and match this behavior with the information gained from direct methods.

SCOPE

The following are discussed in this paper by examining the trajectories of a power system subjected to disturbances such as faults.

- Validity of the concept of a controlling u.e.p. for a particular system transient.
- The manner in which some generators tend to lose synchronism with the rest of the system.
- Analysis of the system energy and defining its components that are directly responsible for system separation.

A VIEW OF TRANSIENT STABILITY

A faulted power system, at the instant of fault clearing, possesses an excess energy that must be absorbed by the network for stability to be maintained. This energy will be referred to here as transient energy. This energy is responsible for setting the synchronous machines to "swing" away from equilibrium, and in the process the transient energy is converted to kinetic energy. For synchronism not to be lost, the system must be capable of absorbing this energy at a time when the forces on the machines tend to bring them back toward new equilibrium positions.

The power system's ability to absorb excess energy depends largely on its ability to convert that energy to potential energy. This in turn depends mostly on the post disturbance network configuration. For a given system configuration, there is a maximum or critical amount of transient energy that the network can absorb and convert to other forms of energy. If the system started with an amount of transient energy less than this critical energy, the machine rotors will swing as far as the system requires for the excess energy to be absorbed by the network. In this case, the system will be stable.

This simple qualitative picture is the basis for many attempts to make an analysis of power system transient stability using direct methods. Quantitative description of the system energies, particularly the critical energy, has been the subject of extensive investigations for many years [1].

Transient Kinetic Energy and the Inertial Center

One fundamental step in defining the energy contributing to system separation is the so-called inertial center formulation of the system equations (also referred to as center-of-angle). The equations describing the behavior of the synchronous machines are formulated with respect to a fictitious inertial center (in contrast to the usual situation where the machine's equations are formulated with respect to a synchronously

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moving frame of reference). The importance of this formulation is in clearly focusing on the motion that tends to separate one or more machines from the rest of the system, and in removing a substantial component of the system transient energy that does not contribute to instability, namely, the energy that accelerates the inertial center [8,9]. With this formulation, the forces tending to separate some machines from the rest of the system, and the energy components associated with their motion, can be easily identified [4,5,9].

Potential Energy Surfaces and the Critical Energy

The following simplified picture emerges from recent research [5,10]. The ability of the system to absorb (or convert) the energy component that contributes to instability depends upon the following: 1) the potential energy contours or "terrain" of the post disturbance system, and 2) the particular segment of this terrain traversed by the faulted trajectory. The former depends upon using a good mathematical accounting of the system energy, which describes the energy surfaces encountered by the machine rotors as they swing away from their equilibrium positions. The energy terrain, as reflected in the potential energy contours, accounts for the amount of the rotor displacement per unit of fault energy resulting from the disturbance. The second point above simply recognizes that those energy surfaces have higher ridges in some segments than in others, and thus, the amount of rotor motion (and the corresponding energy absorbed) necessary to reach instability will vary from one trajectory to another. If the system trajectory moves in a segment of higher potential energy values, the network's capacity to absorb (and convert) the initial excess transient energy is greater, and hence, it can withstand a greater initial disturbance. On the other hand, if the faulted trajectory moves in a region where the potential energy surfaces are "shallow," the network's ability to absorb excess transient energy is much reduced, and instability occurs with a smaller disturbance. Thus, the faulted trajectory is analogous to a particle "climbing up" the potential energy "hills" around this valley. In some directions the ridge, or the peak of the hill, is higher than others.

The ridge of the potential energy surface contours is called the Principal Energy Boundary Surface (PEBS) [5,10]. This ridge has several "humps" and "saddle points." These are the so-called u.e.p.'s, which are connected by the PEBS.

For the first swing transient, the u.e.p. closest to the trajectory of the disturbed system is the one that decides the transient stability of the system. This is called the controlling (or relevant) u.e.p. for this trajectory. Thus, the critical transient system energy is that which corresponds to the energy of the closest or controlling u.e.p. To complete the picture, we mention that if the system is faulted and the fault is cleared before the critical clearing time, t_{cr}, the system trajectory peaks before reaching the "ridge" of the potential energy surface contours or the relevant u.e.p. At a clearing time exceeding t_{cr}, the ridge is crossed (usually at some point other than the u.e.p.) and stability is lost. There is only one critical trajectory that can actually go through the controlling u.e.p.

SYSTEM TRANSIENT ENERGY

In the following discussion, the mathematical model describing the transient power system behavior is the classical model: generators represented by constant voltage behind transient reactance, constant impedance loads, etc. (see Chapter 2 of [11]).

For an n-generator system with rotor angles δ_i and inertia constants M_i, $i = 1,2,\ldots,n$, the position δ_o and speed ω_o of the inertial center are given by

$$\delta_o = \frac{1}{M_o} \sum_{i=1}^{n} M_i \delta_i$$

$$\dot{\delta}_o = \frac{1}{M_o} \sum_{i=1}^{n} M_i \dot{\delta}_i \qquad (1)$$

where

$$M_o = \sum_{i=1}^{n} M_i$$

The generators' angles and speeds with respect to the inertial center are defined by

$$\left.\begin{aligned} \theta_i &= \delta_i - \delta_o \\ \omega_i &= \dot{\delta}_i - \dot{\delta}_o \end{aligned}\right\} \quad i = 1,2,\ldots,n \qquad (2)$$

Using the common terminology

E_i = voltage behind transient reactances of generator i

$Y_{ii} = G_{ii} + jB_{ii}$
= driving point admittance for internal node of generator i

$Y_{ij} = G_{ij} + jB_{ij}$
= transfer admittance between internal nodes i and j

P_{mi} = mechanical power of generator i (constant)

We can show that the system transient energy V is given by

$$V = \sum_{i=1}^{n} \frac{1}{2} M_i \omega_i^2 - \sum_{i=1}^{n} P_i\left(\theta_i - \theta_i^s\right) - \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} \left[C_{ij}\left(\cos\theta_{ij} - \cos\theta_{ij}^s\right) + \int_{\theta_i^s+\theta_j^s}^{\theta_i+\theta_j} D_{ij} \cos\theta_{ij} \, d(\theta_i + \theta_j) \right] \qquad (3)$$

where

θ_i^s = stable equilibrium angle for generator i

$P_i = P_{mi} - E_i^2 G_{ii}$

$C_{ij}, D_{ij} = E_i E_i B_{ij},\ E_i E_j G_{ij}$

The system transient energy components in Eq. (3) are identifiable. The first term is the kinetic energy. The second term is a position energy, which is part of the system's potential energy. The third term is the magnetic energy, which is also part of the potential energy. The fourth term is the dissipation energy, which is the energy dissipated in the network transfer conductances (which includes part of the load impedances). As is common in the literature, we will use the term "potential energy" to indicate the last three components.

Examining Eq. (3) we note that at θ^s the transient energy is zero; and at the instant of fault clearing the transient energy is greater than zero. If the system is to remain stable, the kinetic energy at the beginning of the post disturbance period must be converted, at various instants along the trajectory, to other forms of energy. Thus, the excess (transient) kinetic energy must be absorbed by the network.

It is to be noted that the last term in Eq. (3) can be calculated only if the system trajectory is known (which the direct method is trying to avoid in the first place). Various methods for approximating this term have been suggested [4,5,10,12]. The method used in this investigation is that suggested by Athay et al. [4,5]. The expression for the system transient energy function is given by

$$V = \frac{1}{2}\sum_{i=1}^{n} M_i\omega_i^2 - \sum_{i=1}^{n} P_i\left(\theta_i - \theta_i^s\right)$$

$$- \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left[C_{ij}\left(\cos\theta_{ij} - \cos\theta_{ij}^s\right) + I_{ij}\right] \qquad (4)$$

where

I_{ij} = an approximated term to account for the transfer conductances

$$= D_{ij}\frac{\theta_i + \theta_j - \theta_i^s - \theta_j^s}{\theta_{ij} - \theta_{ij}^s}\left(\sin\theta_{ij} - \sin\theta_{ij}^s\right) \qquad (5)$$

For practical considerations, the transient energy is usually evaluated between two points. Since θ^s for the post disturbance network in a disturbed system may differ from that of the original system, it is important to indicate clearly the reference point for the energy expression. For convenience, the notation $V|_{\theta^a}^{\theta^b}$ will be used to indicate that the transient energy is calculated at point b (θ^b,ω^b) with respect to point a (θ^a,ω^a). For example, for a faulted system, at the instant of clearing, $\theta = \theta^c$, $\omega = \omega^c$. The transient energy with respect to θ^{s1} (the prefault θ^s) is given by

$$V_{cl} = V\Big|_{\theta^{s1}}^{\theta^c}$$

$$= \frac{1}{2}\sum_{i=1}^{n} M_i\left(\omega_i^c\right)^2 - \sum_{i=1}^{n} P_i\left(\theta_i^c - \theta_i^{s1}\right)$$

$$- \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left[C_{ij}\left(\cos\theta_{ij}^c - \cos\theta_{ij}^{s1}\right) + I_{ij}^c\right] \qquad (6)$$

Critical Energy

The critical transient energy is associated with the potential energy of the appropriate u.e.p., θ^u, for the particular disturbance under consideration. The critical disturbance is assumed to be controlled by this θ^u. The system trajectory is assumed, for the time being, to be reaching θ^u with zero velocities. In this case, the critical energy is the same as the potential energy at θ^u, calculated with respect to the post fault equilibrium point θ^{s2}. Thus,

$$V_{cr} = V_u = V\Big|_{\theta^{s2}}^{\theta^u}$$

$$= -\sum_{i=1}^{n} P_i\left(\theta_i^u - \theta_i^{s2}\right)$$

$$- \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\left[C_{ij}\left(\cos\theta_{ij}^u - \cos\theta_{ij}^{s2}\right) + I_{ij}^u\right] \qquad (7)$$

A point of practical significance is to note that V_{cl} and V_{cr} are computed with respect to different reference points.

The unstable equilibrium point

Correct determination of the u.e.p. is essential for the successful use of direct methods, based on Eq. (4), for stability analysis. The method used is based on a package of computer programs supplied to the authors by System Control Inc., and documented in [5]. For the purpose of this investigation, however, it was deemed very important to make absolutely certain that the correct u.e.p. among the numerous possibilities has been found in each case. For this reason, the u.e.p. was usually confirmed by obtaining the swing curves for a near critical disturbance. The appropriate angles from these curves were then used in a Davidon-Fletcher-Powell program to obtain the u.e.p. A similar procedure is used to obtain the post disturbance equilibrium angles, θ^{s2}. From the values of θ^u and θ^{s2}, the energy at θ^u and its components are calculated using Eq. (7).

SIMULATION STUDIES

The Networks Investigated

Two power networks have been used for simulation studies:

1. A 4-generator test system comprising 11 buses and three loads. It is an expanded version of the well-known WSCC 3-generator test system. A one-line diagram of the system is shown in Fig. 1, while generator data and operating conditions are given in the Appendix.

   Three-phase faults near generator No. 4, usually cleared by opening one of the lines 8-10, were investigated.
2. A 17-generator, 163-bus system, which is a reduced version of the network of the State of Iowa. In this version, the study area network and generation are retained, except for combining two generators at the same bus in the Neal plants and at Council Bluffs. The study area contains 11 (actual) generators. The remaining part of the network is replaced with equivalents suited for first swing transient studies, using large but less than infinite inertias. The outside area contains six generators. This power network is shown in Fig. 2, and its data is given in the Appendix.

   This system was tested for first swing transients against the original (and much larger) network used for generation planning studies. Although in the original network the generator models used included field and excitation effects, while in the reduced system the classical model is used, the first swing transient in both cases compared very well.

   Three-phase faults at bus 372 (RAUN), cleared by opening the line 372-193 (to Lakefield), were investigated. The initial condi-

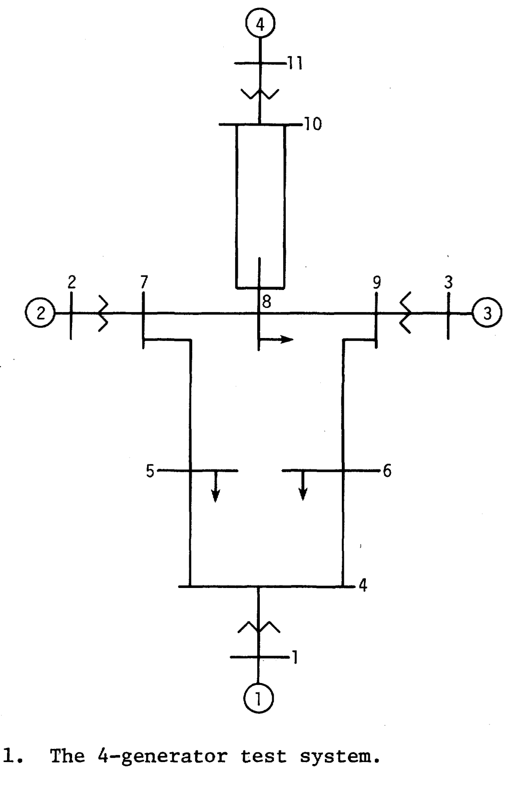

Fig. 1. The 4-generator test system.

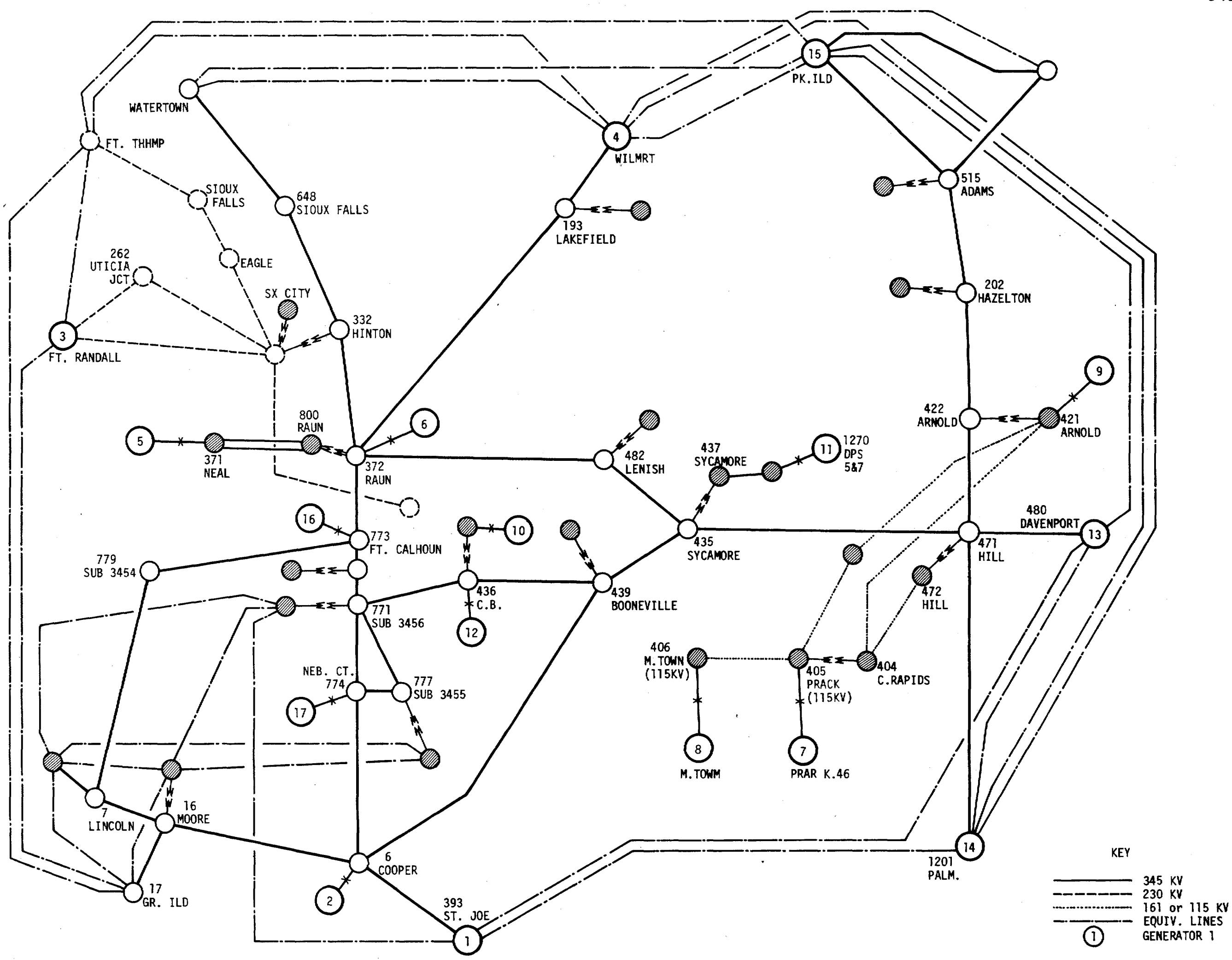

Fig. 2. 17-generator system (Reduced Iowa System)

tion (pre-fault) is for 80% load and prior outage of line 372-332 (RAUN-Hinton).

Unstable Equilibrium Points

For the above mentioned disturbances, the relevant u.e.p.'s were carefully calculated 1) by the special computer program package, and 2) by starting the Davidon-Fletcher-Powell (DFP) procedure from the point where the critical generators were at the peak of their rotor angle swings and the remaining generators were at their θ^{s2} angles. The predicted u.e.p.'s and their potential energies are given below.

4-generator system

$\theta_1^u = -27.7^o$ $\quad$ $\theta_3^u = -11.3^o$

$\theta_2^u = -5.5^o$ $\quad$ $\theta_4^u = 113.2^o$

$V_u = 0.626$ pu

17-generator system

$\theta_1 = -1.4^o$ $\quad$ $\theta_7 = -16.0^o$ $\quad$ $\theta_{13} = -25.8^o$

$\theta_2 = 46.6^o$ $\quad$ $\theta_8 = -8.0^o$ $\quad$ $\theta_{14} = -23.6^o$

$\theta_3 = 9.7^o$ $\quad$ $\theta_9 = -6.6^o$ $\quad$ $\theta_{15} = -17.6^o$

$\theta_4 = -24.0^o$ $\quad$ $\theta_{10} = 47.8^o$ $\quad$ $\theta_{16} = 63.6^o$

$\theta_5 = 163.6^o$ $\quad$ $\theta_{11} = 10.3^o$ $\quad$ $\theta_{17} = 50.1^o$

$\theta_6 = 144.9^o$ $\quad$ $\theta_{12} = 49.6^o$

and $V_u = 17.18$ pu

Casual examination of the above data (i.e., from the values of $\theta_i > \pi/2$) reveals which generators tend to separate from the rest of the system for the specific disturbances given. For the 4-generator system it is generator No. 4, and for the 17-generator system it is generators No. 5 and 6. This data is reasonable, since these generators are close to the disturbance.

System Trajectories

Stability runs, using time solutions and different clearing times, were made for the faults indicated above until the system barely went unstable. In addition to the rotor swings, information on the transient energy was obtained at different instants. It should be noted, however, that the energy is calculated with respect to, the pre-fault stable equilibrium θ^{s1}.

4-generator system

Figures 3 and 4 show some of the results obtained. Machine angles (w.r.t. inertial center) as well as the kinetic energy (K.E.) and potential energy (P.E.), are displayed for the case of t_c slightly less than the critical clearing time, and for the case where t_c was such that the system barely became unstable.

Examining Fig. 3 we note that at the peak of the swing of generator No. 4, the system potential energy is maximum and the kinetic energy is minimum (almost zero in this case). This confirms the idea of the conversion of the kinetic energy to potential energy (noting that a portion of that energy is dissipated in the transfer conductances).

Figure 4 shows that when the P.E. is maximum and the K.E. is minimum, $\theta_4 = 112°$ and $\theta_1 = -29°$ which are almost identical to the values predicted for θ_4^u and θ_1^u. However, the values of θ_2 and θ_3 at that instant are 1.9° and −6.1°. They differ from the predicted values of θ_2^u and θ_3^u by a few degrees. The maximum P.E. is about 0.63 pu, and the minimum K.E., occurring at the same instant, is not exactly zero.

The data shows that at critical clearing the critical machine appears to be at the position predicted by θ_4^u, while the other generators are not exactly at their u.e.p. values.

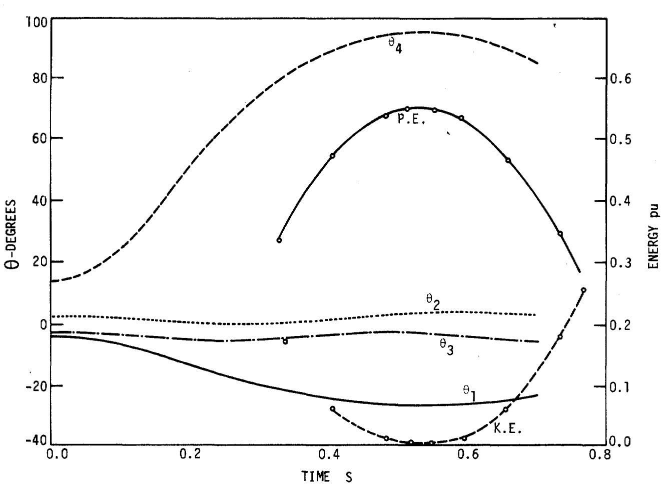

Fig. 3. Four-generator system. Fault at Bus 10 cleared in 0.148s.

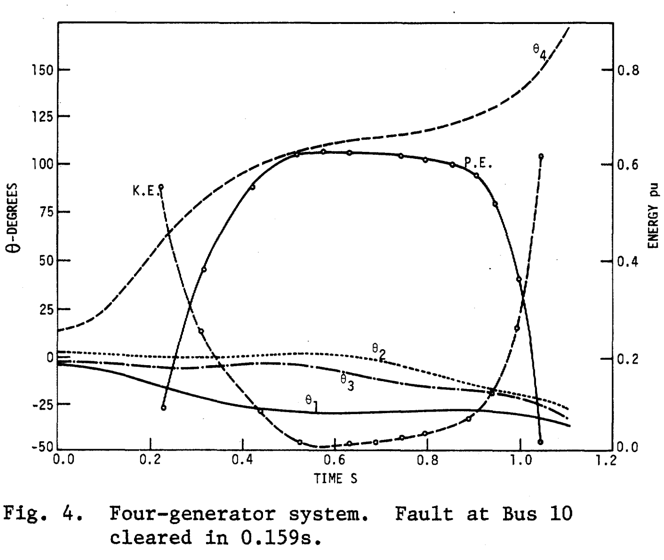

Fig. 4. Four-generator system. Fault at Bus 10 cleared in 0.159s.

17-generator system

One of the important features of this particular system is the nature of the swings of the machines affected by the disturbance, namely, generators No. 5 and 6. Their inertias and their synchronizing forces are substantially different, causing their swings to peak at different instants. Since they represent the generators tending to pull away from the rest of the system, their mode of instability is of interest. To investigate this mode, a series of stability runs was made near the critical clearing time: at $t_c = 0.189$, 0.192, and 0.1932. Plots of θ_5, θ_6, and their inertial center are given in Fig. 5, 6 and 7 together with the system's P.E., K.E., and total energy.

Examining these runs and the figures, we note the following:

- Swings of generators No. 5 and 6 peak about 0.3s apart.
- The peak P.E. and minimum K.E. in Fig. 5 coincide with the peak of the (fictitious) inertial center of the two generators ($\overline{\theta}_{5,6}$).
- For t_c just under the critical value ($t_c = 0.192$s), the peak of $\overline{\theta}_{5,6}$ coincides with a system P.E. nearly equal to that of the u.e.p.; at that instant $\theta_5^u = 156°$ and $\theta_6^u = 141°$, which is very close to the predicted θ_5^u and θ_6^u. The system trajectory continues toward a maximum P.E. and minimum K.E. at a later instant.

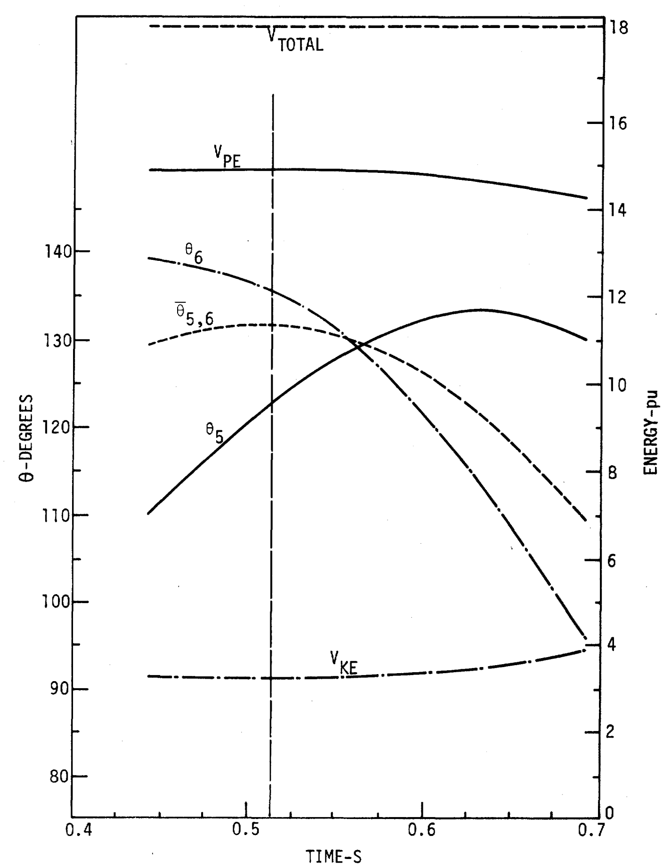

Fig. 5. Seventeen-generator system. Fault at Bus 372 cleared in 0.189s.

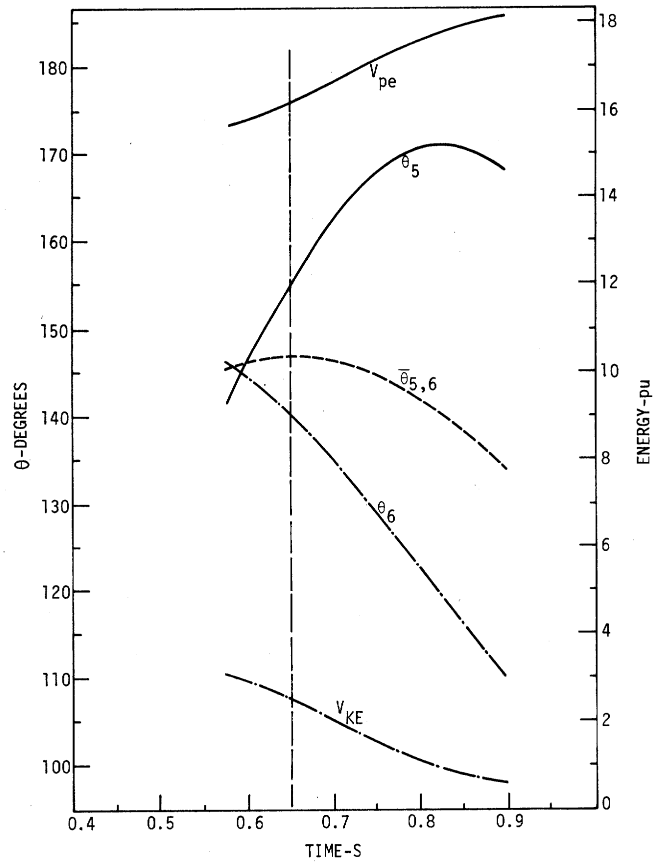

Fig. 6. Seventeen-generator system. Fault at Bus 372 cleared in 0.192s.

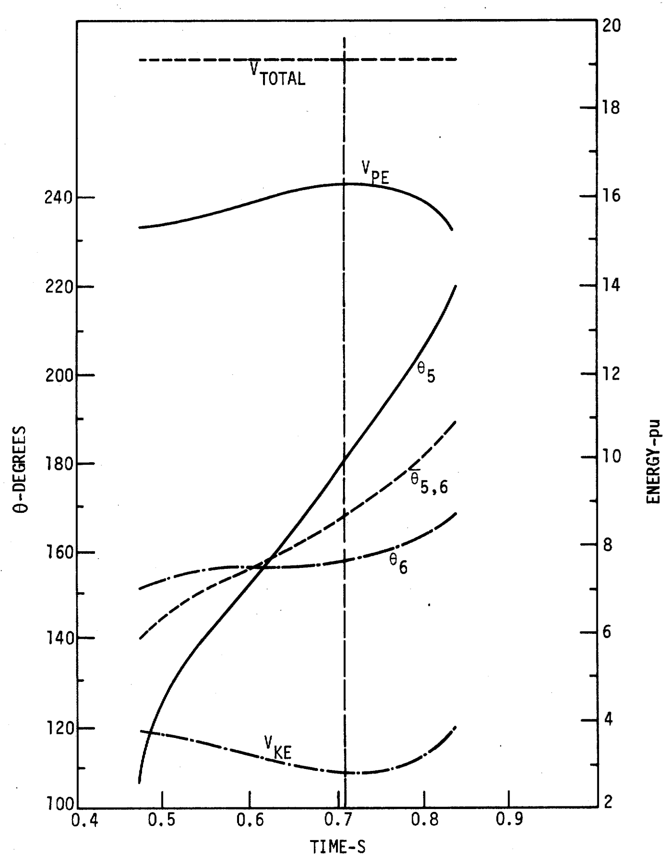

Fig. 7. Seventeen-generator system. Fault at Bus 372 cleared in 0.1932s.

- From the computer runs (not shown in the figures) we note that at the instant of maximum P.E., the rotors of the four equivalent generators, which are remote from the disturbance, are given by

$$\theta_1 = 6.6^o,\ \theta_{13} = -31.5^o,\ \theta_{14} = -30.4^o \text{ and } \theta_{15} = -21.1^o$$

- Therefore, the system trajectory seems to be passing near, but not exactly through, the controlling u.e.p. Again, while the generators tending to separate from the rest of the system pass at or very near their u.e.p. values, the rotors of the other generators are at positions off by a few degrees from their corresponding values at the u.e.p.
- The K.E. minimum is not zero, a point of significance that will be discussed later.
- For $t_c > t_{cr}$ ($t_c = 0.1932$s), the system crossed the potential energy "ridge" at a point different from the u.e.p. At that point, the system P.E. energy is close to that of the u.e.p.

The data presented in Fig. 3-7 shows that the concept of a particular u.e.p. controlling the faulted trajectory is valid. The "critical generators" appear to be at or very near to their θ^u values at critical clearing, and the system P.E., if corrected for the change in θ^s, is very close to that of V_u. However, there are two additional points of significance to be noted: 1) the system minimum K.E. is not zero, and 2) generators other than the critical ones may be off from their θ^u values for trajectories at critical clearing.

Energy Analysis

A more detailed examination of the transient energy of the 17-generator system was carried out at various instants along the system trajectory. Again, the same fault (at bus 372, cleared by opening line 372-193) was investigated for the following instants of clearing.

Fault cleared in 0.15s

The system is stable, and the clearing time is well below critical clearing. Data on this disturbance are displayed in Fig. 8a and 8b. In Fig. 8a, the rotor swings of generators No. 5, 6 and their inertial center, $\bar{\theta}_{5,6}$, as well as those of some other selected generators remote from the disturbance (generators No. 2, 10, 13, 16). The transient energy for this case, as well as its four components, are shown in Fig. 8b.

The plot of the total energy clearly shows the accumulation of excess energy up to the instant of clearing (0.15s) to remain constant afterwards. This shows that no additional excess transient energy is injected into the system following clearing. The variation of the P.E., V_{PE}, (position, magnetic, and dissipation) and the K.E., V_{KE}, are of particular interest. Energy is exchanged back and forth between them. The inertial center of

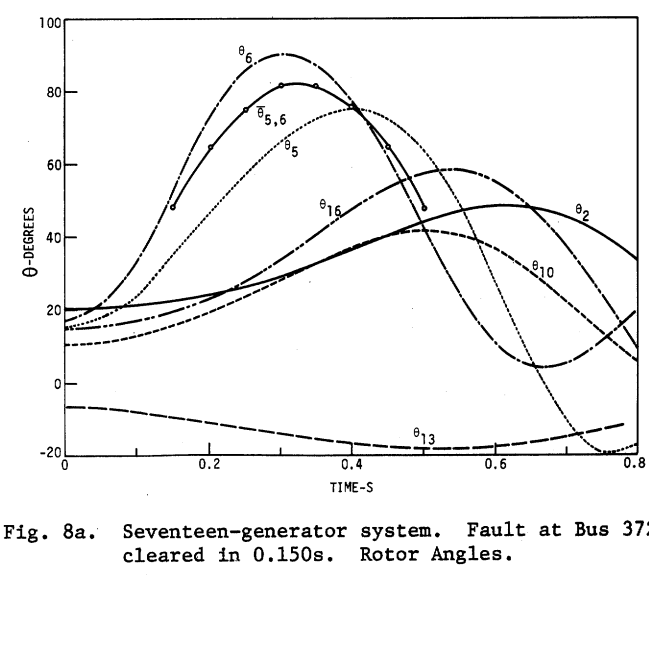

Fig. 8a. Seventeen-generator system. Fault at Bus 372 cleared in 0.150s. Rotor Angles.

the critical generators, i.e., $\bar{\theta}_{5,6}$, acquires zero velocity at the instant at which the P.E., V_{PE}, is maximum and the K.E., V_{KE}, is approximately minimum.

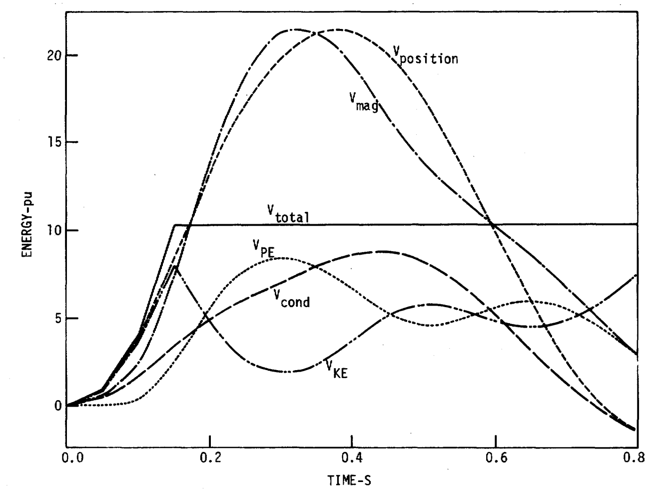

Fig. 8b. Seventeen-generator system. Fault at Bus 372 cleared in 0.150s. Energy.

Fault cleared at 0.1923s, and 0.1926s

These two cases represent critical trajectories. With $t_c = 0.1923$s, the system is critically stable, while with $t_c = 0.1926$s the system is critically unstable. The data for the former case are shown in Fig. 9a (rotor trajectories) and 9b (energy), and the data for the latter case are displayed in Fig. 10a and 10b.

We note that the trajectories pass very near the u.e.p. ($\theta_5 = 166°$, $\theta_6 = 144°$) at about 0.7s. This also corresponds to the instant when $\bar{\theta}_{5,6}$ reaches its peak (or zero velocity) in the stable case, and its inflection point in the unstable case. Thus, we see clearly that the critical generators coincide with their u.e.p. angles at the peak of the swing of their inertial center.

From simple interpolation of the stable trajectory data, the point where $\dot{\bar{\theta}}_{5,6} = 0$ occurs at t = 0.7s and V_{PE} = 16.73 pu. For the unstable trajectory, the relevant point is indicated by the point of inflection, which occurs at t = 0.735s and V_{PE} = 16.97 pu. We conclude from this data that the maximum transient energy that the system can absorb (i.e. the critical energy, V_{cr}) must lie between these two values, or

$$16.73 \leq V_{cr} \leq 16.97$$

Figure 9b and 10b reveal a very important aspect of the energy distribution along the system trajectory. It is sometimes reported that the critical energy, which is related to the network's maximum ability to absorb the excess transient energy, is the maximum P.E. along the trajectory. We

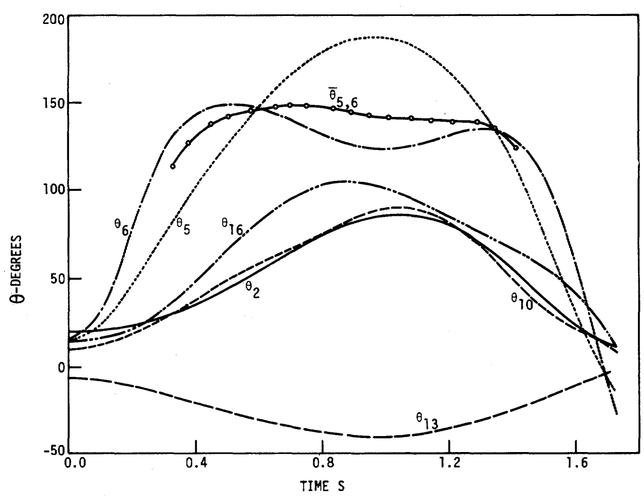

Fig. 9a. Seventeen-generator system. Fault at Bus 372 cleared in 0.1923s. Rotor Angles.

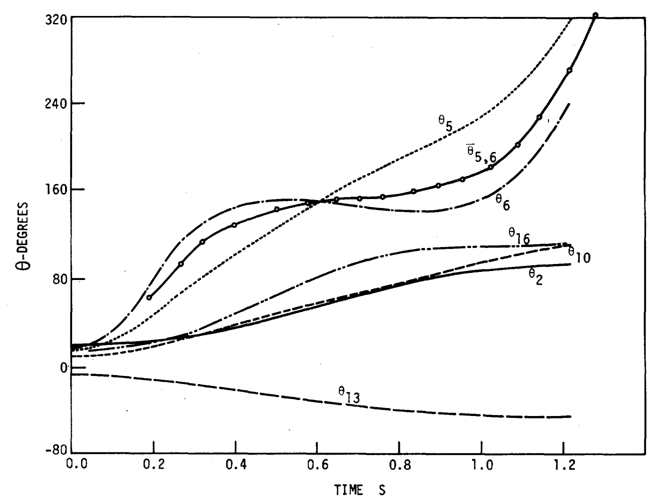

Fig. 10a. Seventeen-generator system. Fault at Bus 372 cleared in 0.1926s. Rotor Angles.

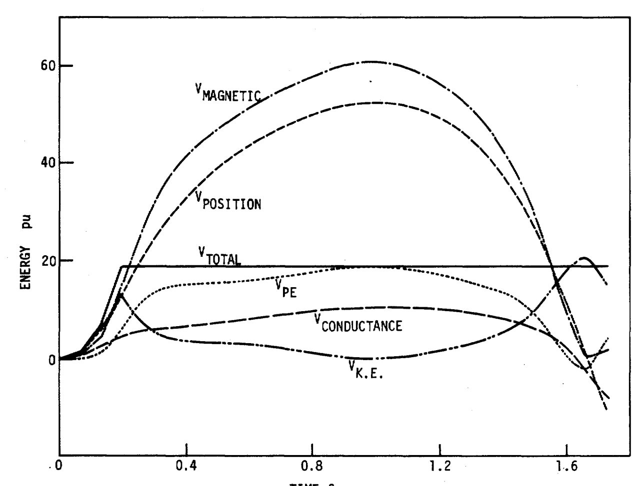

Fig. 9b. Seventeen-generator system. Fault at Bus 372 cleared in 0.1923s. Energy.

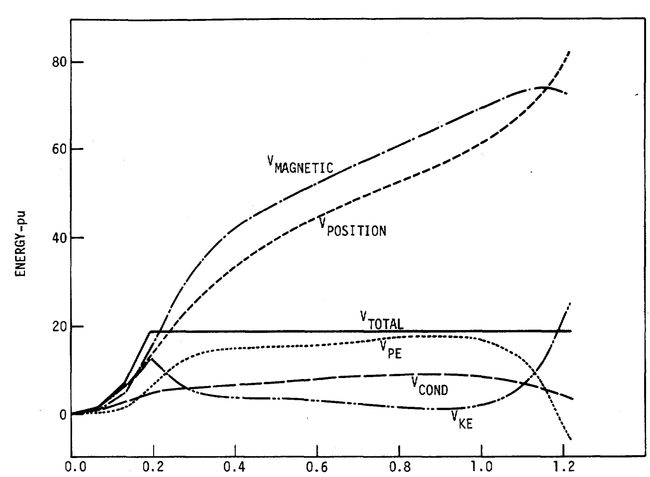

Fig. 10b. Seventeen-generator system. Fault at Bus 372 cleared in 0.1926s. Energy.

have already seen that the critical energy V_{cr} is less than 17 units. Figure 9b shows, however, that the maximum V_{PE} is 18.75 pu and occurs at t = 0.96s. That the system cannot absorb this amount of excess energy is evident from Fig. 10b, where the system becomes unstable when less than this amount of transient energy is injected into the system. Indeed, for that case, when the energy is 17.9 pu (at t = 0.9s), $\overline{\theta}_{5,6}$ is already accelerating toward instability.

An additional observation concerning the K.E. is in order. The instant when the trajectory comes close to the u.e.p., and when $\overline{\theta}_{5,6}$ reaches its peak, the K.E. of the system is not zero; rather, at that instant V_{KE} = 2.17 pu. In the unstable trajectory, the instant at which $\overline{\theta}_{5,6}$ passes the inflection point V_{KE} = 1.967.

The above data are significant for transient stability analysis by direct methods. They clearly indicate the following.

1. The critical transient energy occurs when the critical generators in the system (i.e., the generators tending to separate from the rest) pass at (or very near to) their value at the u.e.p. This amount of critical transient energy appears to be exactly the same as the value of V at the u.e.p. Our own extensive investigations indicate that for all practical purposes the value of V_u (at θ^u) can be used, with sufficient accuracy, as V_{cr}.

2. A certain amount of K.E., between 1.967 and 2.17 units, is not absorbed by the system at the u.e.p. This indicates that not all the excess K.E., created by the fault, contributes to the instability of the system. This component is responsible for much of the inter-generator motion, and not in separating the critical generators from the rest.

DISCUSSION AND CONCLUSIONS

The results of the previous sections merit the following conclusions:

1. The concept of a controlling u.e.p. for a particular system trajectory is valid.
2. From the values of θ_i^u, i = 1,2,...,n, the critical generators, i.e., those tending to separate from the rest, are identified by $\theta_i > \pi/2$.
3. At critical clearing the system trajectory is such that only the critical generators need pass at, or very near to, their values at the u.e.p. Other generators may be slightly off from their u.e.p. values.
4. If more than one generator tends to lose synchronism, instability is determined by the gross motion of these generators, i.e., by the motion of their center of inertia.
5. The value of V_u (at the u.e.p.) is, for all practical purposes, equal to the critical energy, V_{cr}, for the system.
6. Not all the excess K.E. (at t_c) contributes directly to the separation of the critical generators from the rest of the system; some of that energy accounts for the other inter-generator swings. For stability analysis, that component of K.E. should be subtracted from the energy that needs to be absorbed by the system for stability to be maintained.
7. First swing transient stability analysis can be accurately made directly (without time solutions) if:
   - the transient energy is calculated at the end of the disturbance, and corrected for the K.E. that does not contribute to system separation; and
   - the unstable equilibrium point (θ^u) and its energy are computed.

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APPENDIX

Generator data and initial conditions.

Generator No.	Generator Parameters* H MWs/MVA	Generator Parameters* x_d' pu	Initial Conditions P_{mo}^* pu	Internal Voltage E pu	Internal Voltage δ deg.	θ^{s1} deg.
4-generator system						
1	23.64	0.0608	2.269	1.0967	6.95	-4.08
2	6.40	0.1198	1.600	1.1019	13.49	2.45
3	3.01	0.1813	1.000	1.1125	8.21	-2.76
4	6.40	0.1198	1.600	1.0741	24.90	13.91
17-generator system						
1	100.00	0.004	20.000	1.0032	-27.92	-6.26
2	34.56	0.0437	7.940	1.1333	-1.37	20.28
3	80.00	0.0100	15.000	1.0301	-16.28	5.38
4	80.00	0.0050	15.000	1.0008	-26.09	-4.42
5	16.79	0.0507	4.470	1.0678	-6.24	15.41
6	32.49	0.0206	10.550	1.0505	-4.56	17.10
7	6.65	0.1131	1.309	1.0163	-23.02	-1.35
8	2.66	0.3115	0.820	1.1235	-26.95	-5.29
9	29.60	0.0535	5.517	1.1195	-12.41	9.25
10	5.00	0.1770	1.310	1.0652	-11.12	10.53
11	11.31	0.1049	1.730	1.0777	-24.30	-2.64
12	19.79	0.0297	6.200	1.0609	-10.10	11.55
13	200.00	0.0020	25.709	1.0103	-28.10	-6.44
14	200.00	0.0020	23.875	1.0206	-26.76	-5.10
15	100.00	0.0040	24.670	1.0182	-21.09	0.56
16	28.60	0.0559	4.550	1.1243	-6.70	14.95
17	20.66	0.0544	5.750	1.116	-4.35	17.30

*To 100 MVA base.

For Combined discussion see page 3424.